Python-Based Signal Processing Pipeline for PPG Analysis Across Frailty States

Photoplethysmography (PPG) is a non-invasive and widely available method to measure cardiovascular dynamics and is increasingly embedded in wearable technologies. As frailty is associated with vascular stiffness and autonomic dysregulation, PPG holds potential as a low-cost, scalable biomarker for frailty detection. However, most available processing pipelines rely on opaque third-party toolkits, limiting transparency, reproducibility, and clinical adaptability.

This thesis focuses on the design and implementation of a custom Python-based signal processing pipeline to analyze raw PPG signals across frailty states (frail, pre-frail, and non-frail), without the use of external toolkits for core signal analysis. Students will extract time-domain, frequency-domain, and morphological features manually through implemented algorithms, benchmark classification accuracy, and provide insights into physiological differences across frailty stages.

What you do with us

- Gain a deep understanding of cardiovascular biomarkers of frailty, including how they manifest in PPG signals.
- Design and implement preprocessing techniques from scratch (e.g., filtering, peak detection, artifact removal).
- Extract and analyze features related to pulse wave morphology, inter-beat intervals, and variability.
- Develop and evaluate classification models to differentiate frailty states.
- Compare and validate your pipeline against existing reference implementations (e.g., NeuroKit2, HeartPy) for inspiration only.
- Contribute to a transparent, modular Python codebase suitable for integration in wearable frailty monitoring systems.

Applicant profile

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Deliverables

- Fully commented Python codebase (no reliance on third-party toolkits for processing)
- Benchmark results across frailty states
- Detailed documentation of algorithm design and performance evaluation
- Recommendations for integration in wearable health monitoring systems

Notes

- You can write your thesis in English or German.
- You must be able to meet with the AHMS FG once a week to present updates on progress and receive mentoring.

Anticipated duration

6 months

Contact

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Please include a cover letter explaining your background and why you are interested in this thesis topic. In your letter, also list any relevant experience, coursework, or projects related to signal processing, health technology, or machine learning.